

FEM Project 2

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1 Introduction

ANSYS software and tools were used to minimize the weight of a section of a rotating blade assembly under a series of constraints. Response surface optimizations were used on a series of criteria, with use of level-set based topology optimization. Modal analyses were subsequently performed on the disc section at varying rotation speeds.

2 Modeling and Optimization Procedures

All simulations used a 15 degree sector, with fixed supports on the central pipe, and frictionless support boundary conditions on the symmetry condition. A rotational velocity was exerted on the y-axis of 4500 rpm. A custom version of aluminum was used with a young's modulus of 10.5E6 psi, a density of 0.095 *lbs/in*³, and a Poisson's ratio of 0.34. The safety factor requirement of 2 was checked by ensuring a von-Mises stress less than 17500 psi (half of the 35000 psi yield-strength).

2.1 Optimization of Blade and Topology

Optimization of the blade furthest from the axis of rotation was performed first. The project description left five possible constraints for this section - base length and width, end length and width, and chamfer radius. These variables were parameterized and input into two response surface optimizations - one optimizing for weight, and one optimizing for displacement. Results of these analyses can be seen below in figure 1.

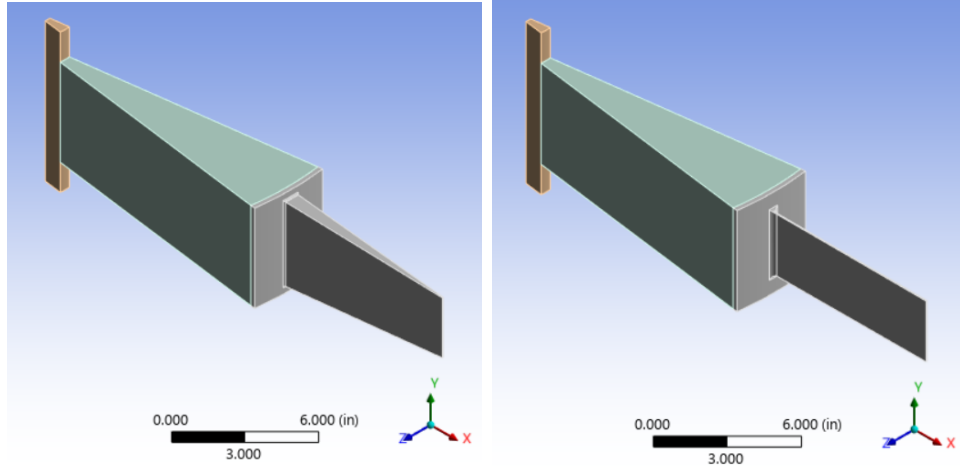


Figure 1: Blade optimization results. Left optimized for displacement. Right optimized for weight

A subsequent level-set-based topology optimization was performed on the section between the blade and the pipe, which will be referred to the midsection. This optimization was used as an assistance for parameterization. The results can be seen in figure 2

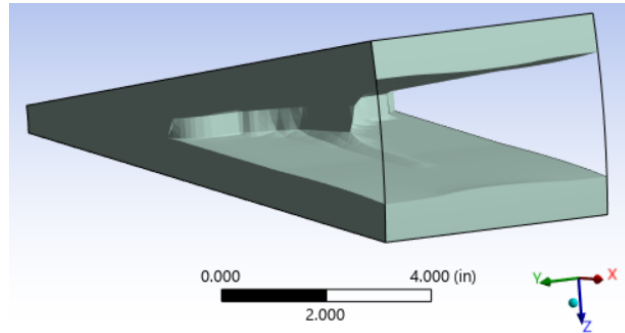


Figure 2: Topology Optimization Results

Both steps as described above were used as assistance in setting up the problem for parameterized response surface optimizations.

2.2 Response Surface Optimizations

A parameterized model was created with a triangular hole in the center of the midsection. The width of the midsection was also reduced and parameterized, also leaving room for chamfers connecting to the pipe and the blade. The blade could be parameterized only using the thickness of the blade at the base - as the tip will always be as small as possible. The parameterized model can be seen below in figure 3.

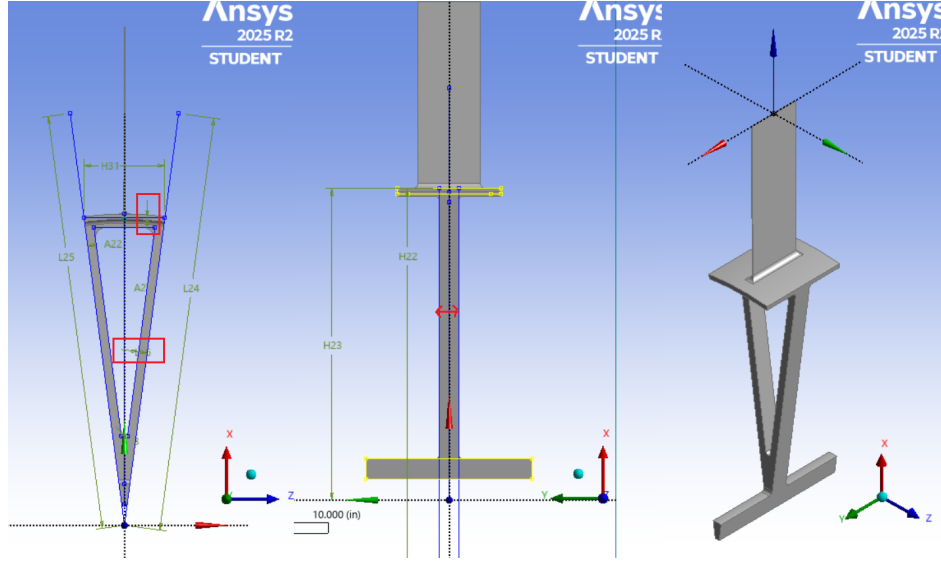


Figure 3: Parameterized Model. From left to right is view from Y, Z, and Iso

Despite efforts to reduce the number of parameters in the problem, there were too many total parameters to run them all in the same simulation in a reasonable time, with a limit on running 4 or 5 parameters at a time. With this in mind, a few optimizations were performed by assuming some of the parameters and discovering which were constrained to their limits. An example of one of these optimizations can be seen in figures 6 and 5 in the appendix. Final calculations were performed with a refined mesh with an element size of 0.15 in, 77252 nodes, and 49675 Elements, which can be seen in the appendix under figure 7.

3 Results and Discussion

The resulting geometry, which can be seen in figure 4, had a mass of 1.9695 lbm, or lbf. The largest radial deflection was 0.014813 inches at 4500 rpm. The largest von-Mises stress in the part at that speed was 17355 psi, translating to a safety factor of 2.017. Plots of these values over the model can be seen in the appendix in figures 8 and 9.

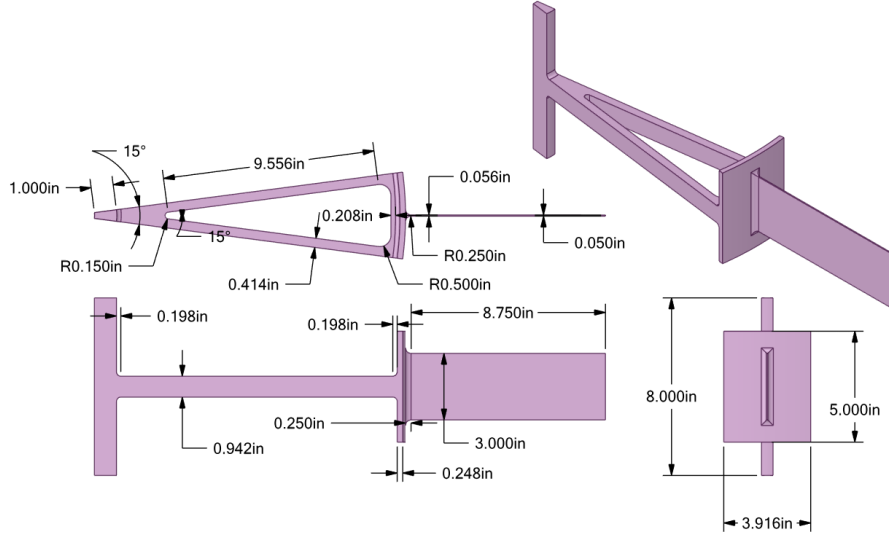


Figure 4: Finalized Geometry of Section

Based on a prestress modal analysis at 4500 rpm, the model's first four undamped natural frequencies were, 126.71 hz, 167.92 hz, 189.67 hz, and 361.97 hz. At an increased speed of 9000 rpm, those frequencies increase to 204.34 hz, 245.5 hz, 278.18 hz, and 666.16 hz. Those frequencies come with a maximum von-Mises stress of 69422 psi, which is above the yield strength of 35000 psi, and a radial deflection of 0.05925 in.

Having explored the entirety of the parametric space under the current configuration of the part, a decently high confidence can be had that it is the best design with that topology. It is highly tapered from the axle outward, with the blade nearly as thin as possible and minimal support around its bracket. Deformation and stress are both active constraints, which tells that there isn't much room for more performance from the part. More time spent exploring the topology of the part could have yielded better results, especially if the support of the midsection was in line with the blade. This is likely the largest flaw in the design.

4 Conclusion

A satisfactory design was achieved using mainly response surface optimization methods, however many mistakes were made during the design process. More time should have been spent exploring the topology of the part at the beginning, because far too many time intensive topology optimizations had to be performed to reach the final result. More time could have been spent on ensuring the parametrized model was sound. Optimizations would often fail because of a bad geometric model. Response Surface-based optimization also appears to be

an inefficient method. It is far too time consuming when using more than 5 parameters. A gradient-based optimization method would be far more efficient with long calculation times, and could incorporate more parameters at once.

5 Appendix

	A	B	C	D	E	F	G	H	I
1	Name	Parameter	Objective		Constraint				
2			Type	Target	Units	Type	Units	Upper Bound	Tolerance
3	Minimize P29	P29 - Geometry Mass	Minimize	0		No Constraint			
4	P27 <= 0.0149 in	P27 - X Axis - Directional Deformation - 1. s Maximum	No Obj...			Values <= Upper Bound		0.0149	0.0001
5	P28 <= 17500 psi	P28 - Equivalent Stress Maximum	No Obj...			Values <= Upper Bound		17500	0.001
*		Select a Para...							

Figure 5: Topology Optimization Setup

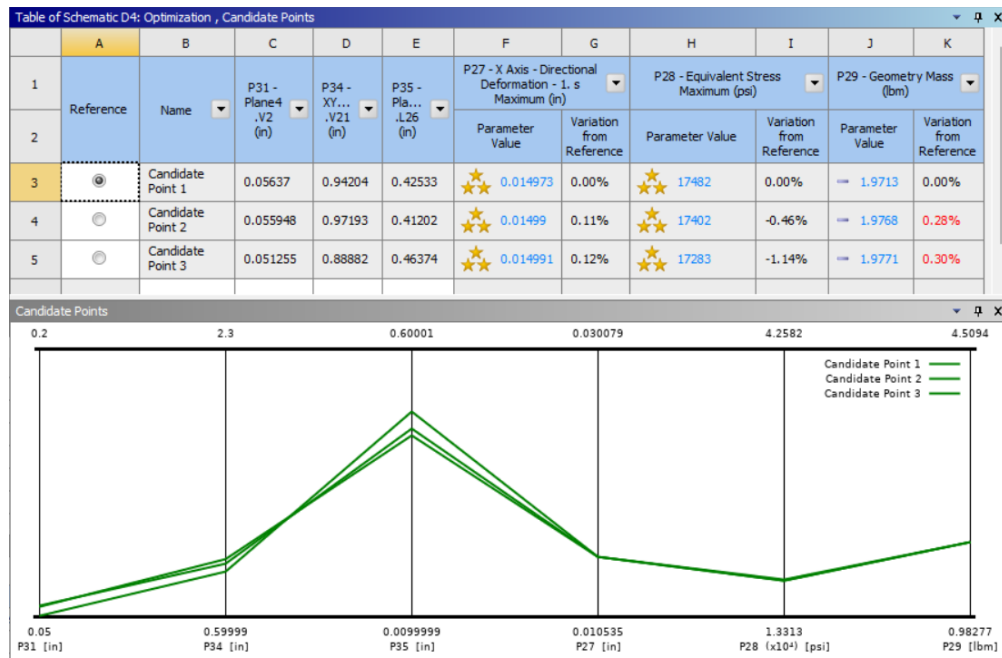


Figure 6: Topology Optimization Results

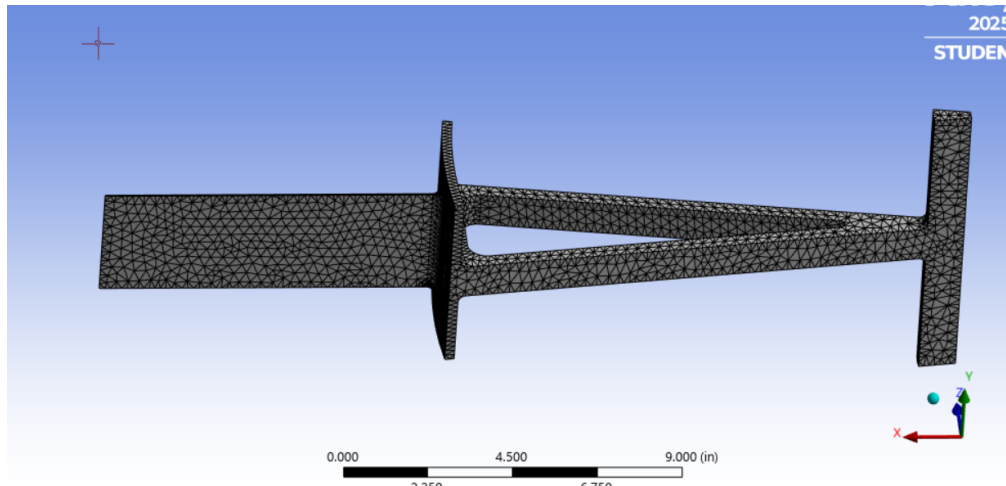


Figure 7: Finalized Mesh

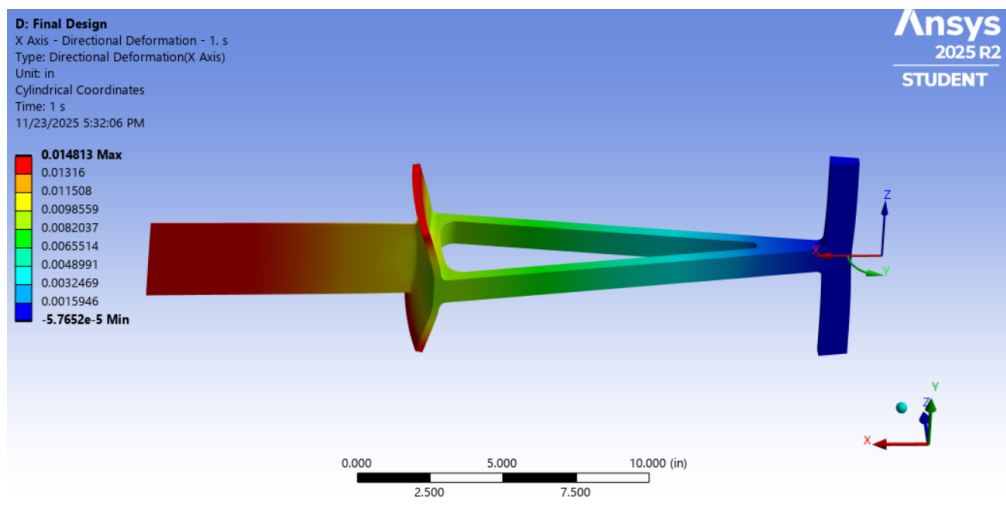


Figure 8: Plot of radial deformation at 4500 rpm

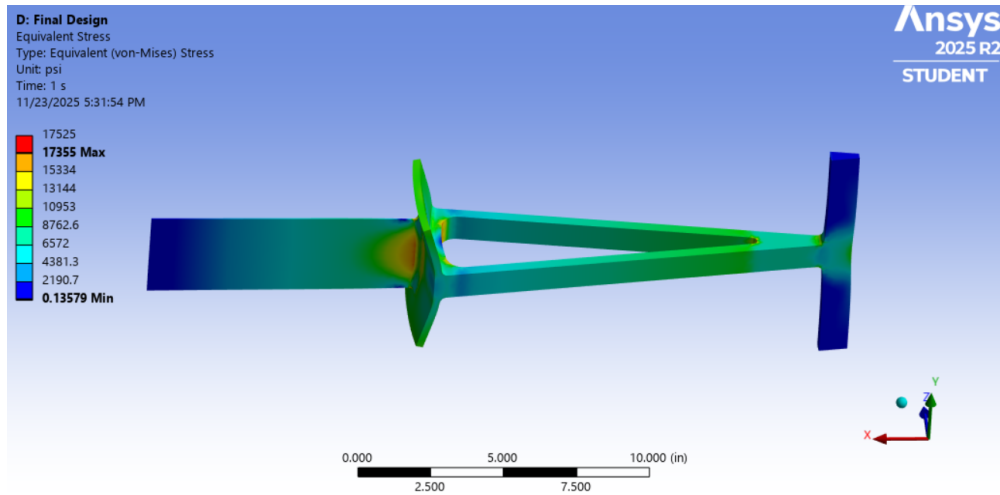


Figure 9: Plot of von-Mises stress at 4500 rpm